

Terrain-Conditioned Cross-Attention for Cross-Region Avalanche Debris Mapping with Sentinel-1 SAR

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Highlights

- The model is tested across four mountain systems and seven avalanche events.
- Three training runs show parity with strong alternatives and a gain only over simple differencing.
- Terrain inputs help in some regions but do not improve accuracy uniformly.
- The slope prior penalizes some pixels that contain genuine avalanche debris.
- The road-corridor test succeeds in the Pamirs but misses the only corridor avalanche in northern Norway.

Author Contributions

Akshay Sharma: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Visualization, Writing—Original Draft.

Lalji Prasad: Supervision, Methodology, Resources, Validation, Writing—Review and Editing.

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Competing Interests

The authors declare no competing financial or non-financial interests.

Ethics Approval and Consent

Not applicable; this study uses public remote-sensing data and involves no human participants or animals.

Consent for Publication

Not applicable.

Data and Code Availability

The AvalCD benchmark is available at <https://doi.org/10.5281/zenodo.15863589>. Code, checkpoints, the JISRS manuscript source, and verification artifacts are archived at <https://github.com/akssha74/TopoRadar-Net-AvalCD/releases/tag/v1.0.23>.

Author Agreement

Both authors approved the manuscript and agree to its submission. The manuscript has not been published and is not under consideration elsewhere.